

Basic Generic Definitions

Example 1 : Generic Identity Function

The identityGB function works for any type:

$[X]$	$\text{identityGB} : X \rightarrow X$
	$\forall x : X \bullet \text{identityGB}(x) = x$

This defines identityGB polymorphically. It can be used with any type: identityGB applied to a natural number returns that number, identityGB applied to a set returns that set, etc.

Example 2 : Generic Constant

A generic empty set:

$[X]$	$\text{empty} : \mathbb{P} X$
	$\text{empty} = \{\}$

The empty set can be instantiated for any type X.

Example 3 : Generic Pair Functions

Extract the first element of a pair:

$[X, Y]$	$\text{fst} : X \times Y \rightarrow X$
	$\forall x : X; y : Y \bullet \text{fst}(x, y) = x$

Extract the second element:

$[X, Y]$	$\text{snd} : X \times Y \rightarrow Y$
	$\forall x : X; y : Y \bullet \text{snd}(x, y) = y$

These work for pairs of any types.

Example 4 : Generic Singleton Set

Create a set containing a single element:

$[X]$	$\text{singleton} : X \rightarrow \mathbb{P} X$
	$\forall x : X \bullet \text{singleton}(x) = \{x\}$

Given any value, singleton produces a set containing just that value.

Example 5 : Generic Sequence Operations

The head of a sequence (generic version):

$$\begin{array}{c} \boxed{\begin{array}{c} [X] \\ \hline \text{headOf} : \text{seq } X \rightarrow X \\ \hline \forall s : \text{seq } X \mid \#s > 0 \bullet \text{headOf}(s) = s(1) \end{array}} \end{array}$$

This is defined only for non-empty sequences.

Example 6 : Generic Set Membership (Conceptual)

Set membership can be tested using the built-in 'in' operator in Z notation. In other contexts, you might define a membership test function, but Z notation provides this as a primitive predicate: $x \text{ in } S$ is true exactly when x is an element of S .

Example 7 : Generic Optional Type (Conceptual)

Optional values can be modeled with free types in Z notation. For a specific type, you would define: $\text{Option} ::= \text{none} \mid \text{some}(T)$. Generic optional types require more complex setup beyond basic gendef blocks, as free types in Z are typically defined for specific types rather than being polymorphic.

Example 8 : Generic Swap Function

Swap the components of a pair:

$$\begin{array}{c} \boxed{\begin{array}{c} [X, Y] \\ \hline \text{swap} : X \times Y \rightarrow Y \times X \\ \hline \forall x : X; y : Y \bullet \text{swap}(x, y) = (y, x) \end{array}} \end{array}$$

Converts (x, y) to (y, x) .

Example 9 : Generic Cardinality

Size of a set (for finite sets):

$$\begin{array}{c} \boxed{\begin{array}{c} [X] \\ \hline \text{size} : \mathbb{P} X \rightarrow \mathbb{N} \\ \hline \forall S : \mathbb{P} X \bullet \text{size}(S) = \#S \end{array}} \end{array}$$

Returns the number of elements in a finite set.

Example 10 : Generic List Construction

Construct a singleton sequence:

$$\begin{array}{c} \boxed{\begin{array}{c} [X] \\ \hline \text{single} : X \rightarrow \text{seq } X \\ \hline \forall x : X \bullet \text{single}(x) = \langle x \rangle \end{array}} \end{array}$$

Wraps a value in a single-element sequence.

Example 11 : Using Generic Definitions

Once defined, generic functions can be instantiated for specific types:

$$\begin{array}{|l} id_nat : \mathbb{N} \longrightarrow \mathbb{N} \\ id_set : \mathbb{P} \mathbb{N} \longrightarrow \mathbb{P} \mathbb{N} \\ \hline id_nat = identityGB[\mathbb{N}] \\ id_set = identityGB[\mathbb{P} \mathbb{N}] \end{array}$$

Here $identityGB[\mathbb{N}]$ is the $identityGB$ function instantiated for natural numbers, and $identityGB[\mathbb{P} \mathbb{N}]$ is instantiated for sets of natural numbers.

Example 12 : Generic Constants

Define a generic empty sequence:

$$\begin{array}{|l} [X] \\ \hline emptySeq : seq \ X \\ \hline emptySeq = \langle \rangle \end{array}$$

This can be instantiated as $emptySeq[\mathbb{N}]$ for an empty sequence of naturals, $emptySeq[Person]$ for an empty sequence of persons, etc.